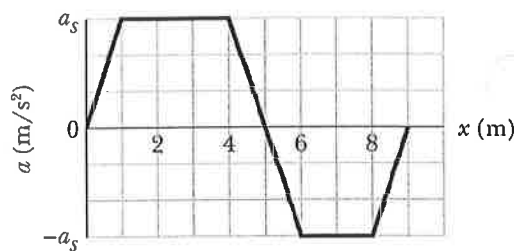


10 points

1. A 4.70kg crate starts from rest at the bottom of a 7.50m long inclined plane. The plane is inclined 20.0° to the horizontal and the crate is dragged up the plane by an applied force of 45.0N and in a direction parallel to the incline. The μ_k between the crate and the incline is 0.27.
 - a) How much work is done by the applied force as the crate moves up the incline?
 - b) How much work is done by gravity?
 - c) How much work is done by friction?
 - d) If the crate starts from rest at the bottom of the incline, how fast is it going at the top?

2. A force $\mathbf{F} = (2.00\mathbf{i} + 9.00\mathbf{j} + 5.30\mathbf{k})\text{N}$ acts on a 2.90kg object in order to make it move from an initial position of $\mathbf{r}_i = (2.70\mathbf{i} - 2.90\mathbf{j} + 5.50\mathbf{k})\text{m}$ to a final position $\mathbf{r}_f = (-4.10\mathbf{i} + 3.30\mathbf{j} + 5.40\mathbf{k})\text{m}$ in 2.10 s.
 - a) How much work is done on the object by the force $\mathbf{F}$ in that time interval?
 - b) What is the average power due to the force during that time interval?
 - c) What is the angle between the vectors $\mathbf{r}_i$ and $\mathbf{r}_f$?

3. A 2.00kg particle starts from rest at $x=0\text{m}$ and an applied force moves it along the x -axis. The figure to the right shows the acceleration of the particle as a function of its position along the x -axis. The scale of the vertical axis is represented by $a_s = 6.0\text{m/s}^2$.



- What is the work done by the force between $x=0$ and $x=4.0\text{m}$?
- What is the work done by the force between $x=4.0\text{m}$ and $x=8.0\text{m}$?
- What is the velocity of the particle at $x=9.0\text{m}$?

4. A trapped miner is rescued from a vertical mine shaft by means of a motor-driven cable. The lift is performed in 10.0m increments. In the first 10.0m, the 80.0kg miner is accelerated from rest to 5.0m/s. In the 2nd 10.0m he is lifted at a constant 5.0m/s. In the 3rd 10.0m he is lifted while decelerating back to zero.

- How much work did the motor do during the first 10.0m? How much power was required?
- How much work did the motor do during the 2nd 10.0m? How much power was required?
- How much work did the motor do during the 3rd 10.0m? How much power was required?